

Assignment 5

Command Line Arguments, File operators, Read and Filters

Write a Bash script that accepts a filename as the first command-line argument.

- Check whether an argument is provided.
- Check whether the file exists using a file operator.
- If the file exists, display the first 5 lines using head.
- Otherwise, display an appropriate error message.

filehead.sh:

```
#!/usr/bin/bash
```

```
if [ -z "$1" ]
```

```
then
```

```
    echo "Error: No filename provided."
```

```
    exit 1
```

```
fi
```

```
if [ -f "$1" ]
```

```
then
```

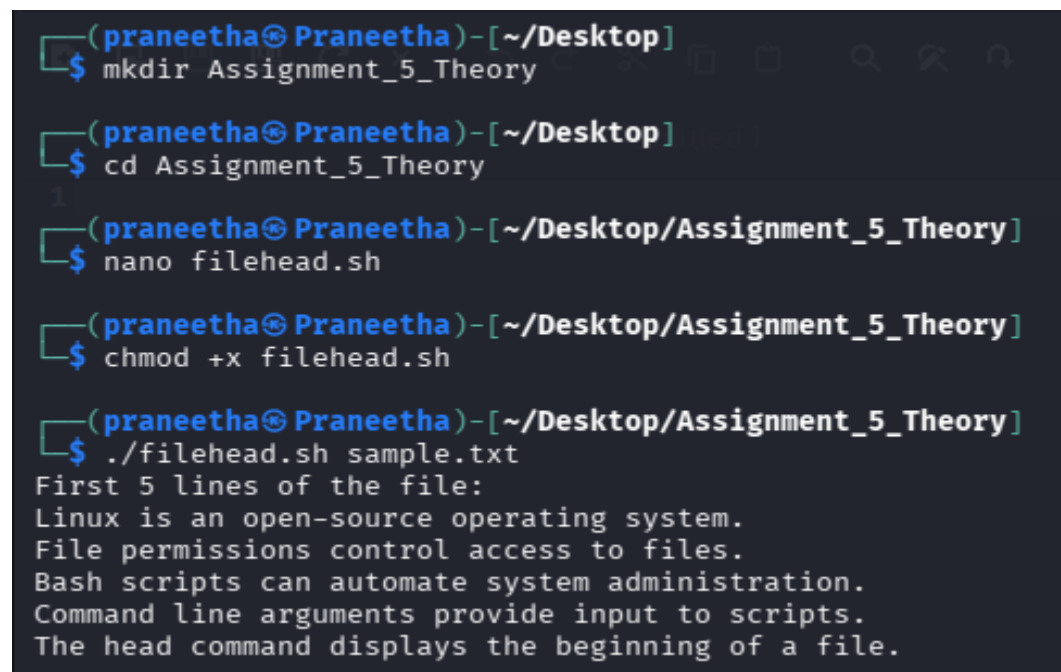
```
    echo "First 5 lines of the file:"
```

```
    head -n 5 "$1"
```

```
else
```

```
    echo "Error: File does not exist."
```

```
fi
```



```
(praneetha@Praneetha)~[~/Desktop]
$ mkdir Assignment_5_Theory

(praneetha@Praneetha)~[~/Desktop]
$ cd Assignment_5_Theory

(praneetha@Praneetha)~[~/Desktop/Assignment_5_Theory]
$ nano filehead.sh

(praneetha@Praneetha)~[~/Desktop/Assignment_5_Theory]
$ chmod +x filehead.sh

(praneetha@Praneetha)~[~/Desktop/Assignment_5_Theory]
$ ./filehead.sh sample.txt
First 5 lines of the file:
Linux is an open-source operating system.
File permissions control access to files.
Bash scripts can automate system administration.
Command line arguments provide input to scripts.
The head command displays the beginning of a file.
```

File Type and Permission Checker

Write a Bash script that accepts a path as a command-line argument and determines whether it is:

- a regular file,
- a directory,
- readable,
- writable, and
- executable.

Display suitable messages using file operators such as `-f`, `-d`, `-r`, `-w`, and `-x`.

checkfile.sh

```
#!/usr/bin/bash
```

```
if [ -z "$1" ]
```

```
then
```

```
    echo "Error: No path provided."
```

```
    exit 1
```

```
fi
```

```
if [ -f "$1" ]
```

```
then
```

```
    echo "$1 is a regular file."
```

```
fi
```

```
if [ -d "$1" ]
```

```
then
```

```
    echo "$1 is a directory."
```

```
fi
```

```
if [ -r "$1" ]
```

```
then
```

```
    echo "$1 is readable."
```

```
else
```

```
    echo "$1 is not readable."
```

```
fi
```

```
if [ -w "$1" ]
```

```
then
```

```
    echo "$1 is writable."
```

```
else
```

```
    echo "$1 is not writable."
```

```

fi
if [ -x "$1" ]
then
    echo "$1 is executable."
else
    echo "$1 is not executable."
fi

```

```

(praneetha@Praneetha)~[~/Desktop/Assignment_5_Theory]
$ nano checkfile.sh

(praneetha@Praneetha)~[~/Desktop/Assignment_5_Theory]
$ chmod +x checkfile.sh

(praneetha@Praneetha)~[~/Desktop/Assignment_5_Theory]
$ ./checkfile.sh sample.txt
sample.txt is a regular file.
sample.txt is readable.
sample.txt is writable.
sample.txt is not executable.

(praneetha@Praneetha)~[~/Desktop/Assignment_5_Theory]
$ ./checkfile.sh myFolder
myFolder is a directory.
myFolder is readable.
myFolder is writable.
myFolder is executable.

```

Write a Bash script that accepts a log file as a command-line argument and searches for the words:

- failed, denied, unauthorized, and error.

```

searchlog.sh

#!/usr/bin/bash

if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -f "$1" ]
then
    echo "Error: Log file does not exist."
    exit 1
fi

grep -Ei "failed|denied|unauthorized|error" "$1"

```

```

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ nano searchlog.sh

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ chmod +x searchlog.sh

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ ./searchlog.sh security.log
ERROR Database connection failed.
WARNING Multiple failed login attempts detected.
ERROR Authentication failed for user charlie.
WARNING Access denied for user david.
ERROR Unauthorized access attempt detected.
ERROR File operation failed.
WARNING Login denied due to invalid password.
ERROR Network connection failed.
WARNING Unauthorized login attempt detected.
ERROR Permission denied while accessing the file.
ERROR Authentication error occurred.
WARNING Failed password attempt detected.

```

Write a Bash script that accepts a filename and checks whether:

- the file exists,
- it is a regular file,
- it is empty or non-empty.
- If it is non-empty, display the first 5 lines.

filecheck.sh

#!/usr/bin/bash

if [-z "\$1"]

then

 echo "Error: No filename provided."

 exit 1

fi

if [-e "\$1"]

then

 echo "File exists."

else

 echo "File does not exist."

 exit 1

fi

if [-f "\$1"]

then

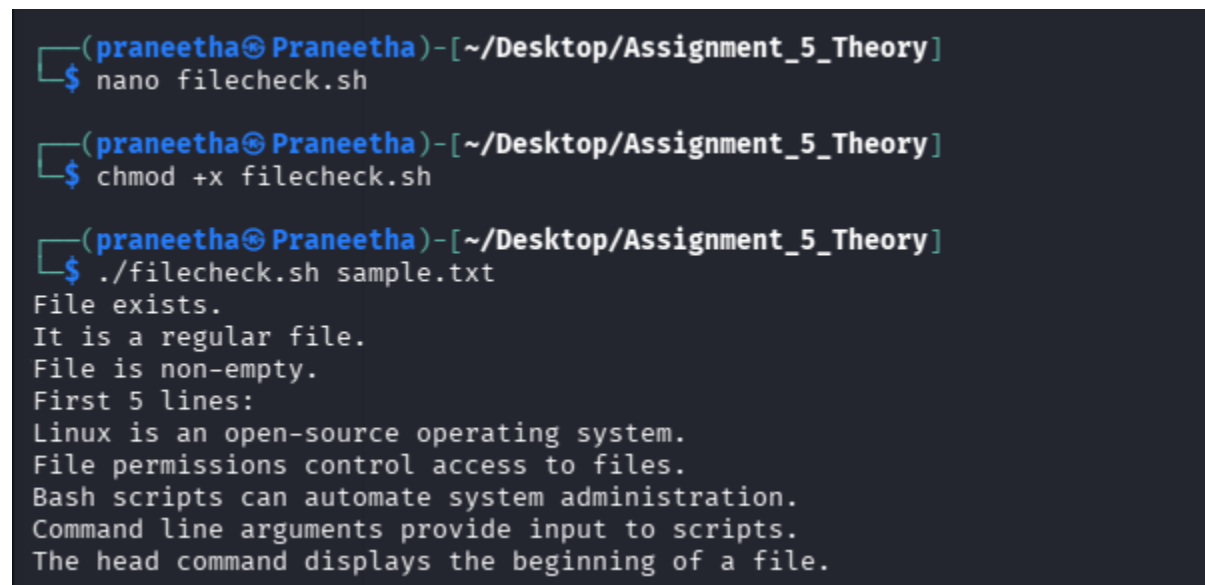
 echo "It is a regular file."

else

```

        echo "It is not a regular file."
    exit 1
fi
if [ -s "$1" ]
then
    echo "File is non-empty."
    echo "First 5 lines:"
    head -n 5 "$1"
else
    echo "File is empty."
fi

```



```

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ nano filecheck.sh

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ chmod +x filecheck.sh

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ ./filecheck.sh sample.txt
File exists.
It is a regular file.
File is non-empty.
First 5 lines:
Linux is an open-source operating system.
File permissions control access to files.
Bash scripts can automate system administration.
Command line arguments provide input to scripts.
The head command displays the beginning of a file.

```

Write a Bash script named investigate.sh that accepts a log file as a command-line argument.

- The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term.
- Display the first 5 matching records and report the total number of matching records.

investigate.sh:

```

#!/usr/bin/bash

if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -e "$1" ]

```

```

then

    echo "Error: File does not exist."

    exit 1

fi

if [ ! -r "$1" ]

then

    echo "Error: File is not readable."

    exit 1

fi

read -p "Enter search term: " term

echo "First 5 matching records:"

grep -i "$term" "$1" | head -n 5

count=$(grep -ic "$term" "$1")

echo "Total matching records: $count"

```

```

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ nano investigate.sh

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ chmod +x investigate.sh

(praneetha@Praneetha)-[~/Desktop/Assignment_5_Theory]
$ ./investigate.sh security.log
Enter search term: failed
First 5 matching records:
ERROR Database connection failed.
WARNING Multiple failed login attempts detected.
ERROR Authentication failed for user charlie.
ERROR File operation failed.
ERROR Network connection failed.
Total matching records: 6

```